

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Civil Engineering Semester : VI Subject Code & Name: BTCVE605E Advanced Soil Mechanics Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	What is the meant by 'Primary valence bond' and 'Secondary valence bond'? Which is more important in soil engineering?	CO1	(06)
B)	Write short note on electric charges on clay minerals.	CO1	(06)
C)	Discuss the characteristic and construction of Kaolinite mineral and Montmorillonite mineral with neat sketch.	CO1	(06)
Q.2	Solve Any Two of the following.		
A)	Derive an expression for vertical stress at a point due to a concentrated load using Boussinesq's theory.	CO1	(06)
B)	A single concentrated load 1000 kN acts at the ground surface. Construct an isobar for a vertical pressure of intensity of 40 kN/sqm. Use Boussinesq's formula	CO2	(06)
C)	Write the advantages and disadvantages of Newmark's influence chart.	CO2	(06)
Q. 3	Solve Any One of the following.		
A)	Enlist equipment used for compaction and explain any one of them in detail with neat sketch.	CO2	(06)
B)	Describe the pre compression techniques of cohesive soil in detail.	CO3	(06)
C)	Explain the additives in soil technique of ground improvement.	CO3	(06)
Q.4	Solve Any Two of the following.		
A)	Explain the procedure of soil reinforcement to slopes of embankment with neat sketch	CO4	(06)
B)	Explain in detail geotextile and its uses.	CO4	(06)

C)	Enlist the method used for ground improvement of cohesion less soil and explain any one of them with neat sketch.	CO4	(06)
Q. 5	Solve Any One of the following.		
A)	Explain in detail the types of grouts in grouting technique of soil.	CO5	(06)
B)	Write the application of displacement compaction grouting technique of soil with neat sketch	CO5	(06)
C)	Write factors affecting selection of construction equipment.	CO5	(06)
	*** End ***		